Wesley Rancher

wesr@uoregon.edu • Personal Website

SUMMARY

Geospatial data analyst trained in remote sensing, GIS, and ecological modeling. Experienced in building repeatable pipelines for scientific and enterprise datasets using Python, R, SQL, and cloud frameworks. Work focuses on automation, application development, and spatial analysis across infrastructure and environmental domains.

EDUCATION

University of Oregon, M.S. in Geography, GPA: 3.9

Ohio Wesleyan University, B.A. in Environmental Studies and Geography, GPA: 3.5

TECHNICAL SKILLS

Languages: Python, R, SQL, Bash

Spatial: ArcGIS Pro/Enterprise, ArcGIS Online, QGIS, Google Earth Engine, Rasterio, GeoPandas, Terra, GDAL

Machine Learning: Scikit-learn, PyTorch, Tidymodels, Random Forest, U-Net, feature engineering

Data Engineering: NumPy, Pandas, Xarray, Docker, Git, REST APIs, AWS CLI

Applications: Remote sensing, change detection, ecological and predictive modeling, spatial analysis

PROFESSIONAL EXPERIENCE

GIS Analyst | American Electric Power | *Columbus, OH* **Oct 2025 – Present**

- Co-maintain and enhance AEP Real Estate's enterprise GIS and SDE datasets across an 11-state territory, supporting property records for owned and leased assets, easements, and other real estate agreements
- Administer and maintain Smartsheet as a core department data system, managing project data, automations, dashboards, reports, and Python-based integrations for real estate workflows
- Implement Python-based GIS automation using Smartsheet SDK, ArcGIS, and ArcPy to connect active real estate projects and transaction information with parcel data and ArcGIS Online applications
- Develop ArcPy and Python tools for IBM TRIRIGA integration, including processing enterprise reports and OSLC/REST data, and improving parcel lookup and selection workflows with GeoParquet; reduced individual parcel selection tasks by approximately 10 minutes
- Improve automated workflows for converting historical property deeds into draft parcel geometries, using OCR and text extraction to identify geographic references, match descriptions to existing GIS data, and convert metes-and-bounds descriptions into geometries with GeoPandas, Shapely, and geometric calculations
- Build computer vision workflows using RGB and NIR drone imagery to identify shoreline infrastructure at a major hydroelectric asset, including image preprocessing, tiling, training-data generation, spectral indices, thresholding, and Random Forest-assisted labeling
- Leverage AI-assisted development with Claude Code to accelerate development and debugging of Python GIS automation, enterprise data integrations, and internal applications
- Mentor new team members on GIS workflows, real estate fundamentals, and automation practices

RESEARCH & TEACHING EXPERIENCE

Graduate Research Assistant | University of Oregon | *Eugene, OR* **Sep 2023 – Aug 2025**

Advisor: Dr. Melissa Lucash

- Collaborated with researchers from the Bonanza Creek Long Term Ecological Research (LTER) program to evaluate wildfire-driven vegetation change and ecosystem dynamics under a changing climate
- Developed an end-to-end remote sensing workflow integrating multispectral time series, Google Earth Engine, Random Forest regression models, and LANDIS-II simulations to quantify aboveground biomass dynamics across eight million hectares of interior Alaska
- Built image preprocessing pipelines, including atmospheric and topographic correction, spectral index calculation, and cross-sensor harmonization
- Containerized the LANDIS-II ecosystem model using Docker to improve reproducibility and simplify deployment of large-scale landscape simulations in cloud environments

Graduate Teaching Assistant — Remote Sensing I & Global Wildfire — **Sep 2023 – Spring 2025**

- Developed lab exercises, taught GIS and remote sensing software (ArcGIS, QGIS, R), and led hands-on demonstrations for undergraduate and graduate student
- Supported curriculum development, supplemental instruction, and student assistance for Global Wildfire
- Delivered guest lectures on changing wildfire in Brazil and the relationship among post-wildfire effects, vegetation, and pollinators

Graduate Research Fellow | NASA DEVELOP | *Virtual* — **Jun 2023 – Aug 2023**

Advisor: Dr. Anthony Vorster

- Developed Random Forest models integrating vegetation indices, LiDAR, and field observations to map invasive vegetation throughout the Paria River watershed
- Processed multispectral imagery and LiDAR using ArcGIS Pro and Google Earth Engine to generate reproducible geospatial products supporting land management decisions
- Presented technical findings to federal and nonprofit partners while co-authoring the final project report

Undergraduate Research Assistant | Ohio Wesleyan University — **Dec 2022 – May 2023**

Advisor: Dr. Nathan Rowley

- Estimated glacial lake depth in western Greenland using radiative transfer methods and Landsat imagery, resulting in a peer-reviewed publication in Glacies
- Refactored a MATLAB-based raster sieving workflow into R, reducing code complexity while simplifying raster-based feature delineation
- Developed reproducible remote sensing workflows for NDWI-based glacial lake mapping and analysis

Undergraduate Research Fellow | University of Central Oklahoma — **Jun 2022 – Jul 2022**

Advisor: Dr. Victor Gonzalez

- Conducted field and laboratory experiments evaluating thermal tolerance of pollinators under environmental stress
- Co-authored a peer-reviewed publication in the Journal of Experimental Biology and presented at the International Union for the Study of Social Insects

SELECTED PUBLICATIONS

- Weiss, S., **Rancher, W.**, Hayes, K., Buma, B., & Lucash, M. (2026). Wildfire dynamics under climate change in interior Alaska. Canadian Journal of Forest Research.
- Lucash, M., et al., including **Rancher, W.** (2026). Roadmap for the future of extreme wildfire events. Fire Ecology.
- Gonzalez, V., **Rancher, W.**, et al. (2024). Bees remain heat tolerant after acute exposure to desiccation and starvation. Journal of Experimental Biology.
- Rowley, N., **Rancher, W.**, & Karmosky, C. (2024). Comparison of multiple methods for supraglacial melt-lake volume estimation in western Greenland during the 2021 summer melt season. Glacies.

SELECTED PRESENTATIONS

- **Rancher, W.** (2025). Estimating species-level aboveground carbon in interior Alaska using machine learning and process-based models. University of Oregon.
- **Rancher, W.**, Matsumoto, H., Lamping, J., & Lucash, M. (2025). Estimating recent shifts in aboveground carbon and species composition in interior Alaska using Landsat imagery and random forests. Northwest Scientific Association.
- **Rancher, W.**, Matsumoto, H., Lamping, J., & Lucash, M. (2024). Assessing vegetation shifts in boreal Alaska by integrating Landsat imagery with spatial modeling. American Geophysical Union.
- **Rancher, W.**, VanArnam, M., Kowalski, A., Anarella, T., & Vorster, A. (2023). Mapping Russian olive and tamarisk to inform invasive species management along the Paria River, Utah. NASA DEVELOP Day.

AWARDS & HONORS

2024 Rippey Research Grant ($1,000), University of Oregon

2023 NASA DEVELOP Scholarship ($1,500), Science Systems and Applications Inc.

2023 Robert E. Shanklin Distinguished Scholar, Geography, Ohio Wesleyan University

2023 Dean’s List; Phi Sigma Tau, Ohio Wesleyan University

2022 Our New Gold Digital Storytelling Winner, Spanish, Ohio Wesleyan University